VR SIDDHARTHA ENGINEERING COLLEGE, VIJAYAWADA
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DefectVision AI: Explainable Multi-Model Industrial Defect Detection and Intelligent Quality Inspection System

23AI&ML7552: B. Tech Mini Project (0th Review)

18th August 2026

Batch No: Non Elite -15

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Abstract

Industrial quality inspection is a critical process in manufacturing, where manual inspection is often **time-consuming, inconsistent, and error-prone**. This project proposes **DefectVision AI**, an **AI-powered intelligent inspection system** for automated defect detection and anomaly localization using deep learning techniques. The system integrates **supervised models (CNN-based classifiers)** and **unsupervised anomaly detection methods (Autoencoders, PatchCore, PaDiM)** to detect defects with high accuracy.

The proposed framework not only identifies defective products but also provides **visual explanations using Grad-CAM and anomaly heatmaps**, improving model interpretability. A **severity scoring mechanism** is introduced to classify defects into minor, moderate, and critical categories based on anomaly intensity and defect area. Additionally, the system supports **batch inspection, folder monitoring, and real-time camera-based inspection**, simulating real industrial environments.

Advanced features such as **similarity-based defect retrieval, active learning, and drift detection** enhance adaptability and continuous improvement. The system is designed as a **scalable smart manufacturing solution**, capable of integration with industrial cameras and automated pipelines. Experimental evaluation on the **MVTec AD dataset** demonstrates strong performance in both detection and localization tasks.

Keywords: Industrial Defect Detection, Computer Vision, Deep Learning, Anomaly Detection, Explainable AI, Smart Manufacturing, Quality Inspection

Presentation Outline

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1. Research Domain

Primary Domain: Artificial Intelligence and Machine Learning

Sub-Domain: Computer Vision and Deep Learning

Application Area: Industrial Automation and Smart Manufacturing

This project focuses on applying **deep learning-based anomaly detection and explainable AI techniques** for real-time industrial quality inspection and defect analysis.

2. SDG Categories

- SDG 9: Industry, Innovation and Infrastructure
  - Promotes development of **smart manufacturing systems**
  - Enhances industrial productivity through AI
- SDG 12: Responsible Consumption and Production
  - Reduces defective products and material wastage
  - Improves quality control processes
- SDG 8: Decent Work and Economic Growth
  - Increases efficiency and reduces manual labor dependency



3. Aim and Motivation

Aim

- To develop an **AI-based intelligent quality inspection system** for accurate defect detection, localization, and severity analysis in industrial products.

Motivation

- Manual inspection is **slow and inconsistent**
- Industries require **automated, scalable, and reliable solutions**
- Need for **explainable AI in critical decision systems**
- Growing demand for **Industry 4.0 and smart manufacturing**

4. Research Questions

- ✓ How can deep learning models be used to accurately detect and localize industrial defects in real-world datasets?
- ✓ How can explainable AI techniques improve transparency and trust in automated defect detection systems?
- ✓ How can anomaly detection and severity scoring be integrated to enhance industrial decision-making and quality control?

5. Title Justification

- "DefectVision AI" represents **vision-based intelligent defect detection**
- "Explainable" highlights **model transparency using Grad-CAM and heatmaps**
- "Multi-Model" reflects integration of **CNN, Autoencoder, PatchCore, and PaDiM**
- "Industrial Defect Detection" specifies the **application domain**
- "Quality Inspection System" emphasizes **real-world usability and deployment capability**

6. Objectives

- To design and implement a deep learning-based system for detecting defective and non-defective products.
- To develop anomaly localization techniques using heatmaps and segmentation for identifying defect regions.
- To integrate explainable AI and severity scoring mechanisms for improved decision-making.
- To simulate real-world industrial inspection using batch processing, folder monitoring, and live camera integration.

7. Scope

- The project focuses on developing an AI-based industrial defect detection system using deep learning models for classification, anomaly localization, and explainable analysis on real-world datasets such as MVTec AD.

- The system includes advanced features such as batch inspection, real-time monitoring, severity scoring, similarity search, and LLM-based defect explanation to simulate real industrial quality inspection workflows.

- The implementation is limited to software-based simulation and does not involve direct integration with industrial hardware, but it is designed to be scalable and adaptable for real-world deployment.

8. Timeline Chart

Table No. : Timeline chart for Mini Project work

S.No	Task	August	September	October	November
1	Literature Survey	✓			
2	Dataset Collection & Analysis	✓	✓		
3	Data Preprocessing		✓		
4	Model Implementation		✓	✓	
5	Model Evaluation			✓	
6	Advanced Features			✓	✓
7	UI & Backend Development				✓
8	Deployment				✓
9	Report Writing			✓	✓
10	Paper Submission				✓

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